# Panel Data

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## I. Introduction

The term *panel data* is used in econometrics to refer to data sets with *repeated observations* over time (or other dimensions) for a given cross-section of units. Units can be persons, households, firms, countries, etc. Observations may be repeated over time (the typical case), but the dimension of repetition can also be something else (e.g. twins: for each unit, a family, we have two observations, the twins).

Panel data are different from *repeated cross-sections* in that the same (identifiable) units are observed at different points in time. In this chapter, we use individuals, denoted by $i$, to refer to units, and time, denoted by $t$, to refer to the different observations we have for a unit.

The main advantages of panel data are two. First, they allow us to control for *permanent unobserved heterogeneity*, i.e. relevant variables that are fixed over time, but unobserved by the econometrician. Second, they allow us to analyze *dynamic responses* and *error components*. The latter refers to decomposing the unobserved term into pieces (e.g. permanent and transitory components) and studying their variance/covariance structure. The former refers to the fact that outcomes and covariates may exhibit persistence and feedback effects (past variables affecting future outcomes).

Over this chapter, we use the following example. Consider the estimation of the demand for cigarettes. Let $C_{it}$ denote the number of cigarettes consumed per day by individual $i$ in year $t$, let $P_{it}$ denote the price at which this person is exposed, and let $Y_{it}$ denote income level. Holding income level constant might not be enough to capture systematic differences across individuals. Different individuals may have different propensity for being a smoker, which might correlate both with the prices they are exposed to and their income. Thus, the final demand for cigarettes can be expressed as:

$$\ln C_{it} = \beta_0 + \beta_1 \ln P_{it} + \beta_2 \ln Y_{it} + \eta_i + v_{it}, \tag{1}$$

where $\eta_i + v_{it}$ is not observed by the econometrician. We interpret $\eta_i$ as an individual-specific, time-invariant component (permanent unobserved heterogeneity), while $v_{it}$ captures idiosyncratic shocks. Throughout the chapter, we compare estimators under different assumptions on whether $\eta_i$ is correlated with regressors.

## II. Static Models

We initially consider the static panel data model. Let us first introduce some general notation. Our model is rewritten as:

$$y_{it} = \boldsymbol{x}_{it}'\boldsymbol{\beta} + (\eta_i + v_{it}), \tag{2}$$

where $y_{it}$ and $\boldsymbol{x}_{it}$ are observed by the econometrician, and $\eta_i$ and $v_{it}$ are not observed. We use subscripts to distinguish individual-specific components and time-varying components, and we use vector/matrix notation to stack observations over time and across individuals. Let $\{y_{it}, \boldsymbol{x}_{it}\}_{i=1,...,N}^{t=1,...,T}$ be our sample. We define $\boldsymbol{y}_i \equiv (y_{i1}, ..., y_{iT})'$, $\boldsymbol{y} \equiv (\boldsymbol{y}_1', ..., \boldsymbol{y}_N')'$, $X_i \equiv (\boldsymbol{x}_{i1}, ..., \boldsymbol{x}_{iT})'$, $X \equiv (X_1', ..., X_N')'$, $\boldsymbol{\eta}_i \equiv \eta_i \iota_T$ where $\iota_T$ is a size $T$ vector of ones, $\boldsymbol{\eta} \equiv (\boldsymbol{\eta}_1', ..., \boldsymbol{\eta}_N')'$, $\boldsymbol{v}_i \equiv (v_{i1}, ..., v_{iT})'$, and $\boldsymbol{v} \equiv (\boldsymbol{v}_1', ..., \boldsymbol{v}_N')'$. Therefore, for our sample, we can rewrite the model as:

$$\boldsymbol{y}_i = X_i\boldsymbol{\beta} + (\boldsymbol{\eta}_i + \boldsymbol{v}_i), \tag{3}$$

for $\{\boldsymbol{y}_i, X_i\}_{i=1}^N$, or:

$$\boldsymbol{y} = X\boldsymbol{\beta} + (\boldsymbol{\eta} + \boldsymbol{v}). \tag{4}$$

Both compact notations are very useful in derivations.

For static models, we maintain the following assumptions:

- *Fixed effects vs random effects:* we allow for two cases, $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] \neq 0$ (fixed effects), or $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] = 0$ (random effects). The distinction determines whether pooled OLS/GLS are consistent.
- *Strict exogeneity:* $\mathbb{E}[\boldsymbol{x}_{it}v_{is}] = 0 \quad \forall s, t$. This rules out feedback from past idiosyncratic shocks into current regressors (e.g. it rules out including lagged dependent variables among $\boldsymbol{x}_{it}$).
- *Error components:* $\mathbb{E}[\eta_i] = \mathbb{E}[v_{it}] = \mathbb{E}[\eta_i v_{it}] = 0$.
- *Serially uncorrelated shocks:* $\mathbb{E}[v_{it}v_{is}] = 0 \quad \forall s \neq t$.
- *Homoskedasticity and independence (for convenience):* $\eta_i \sim iid(0, \sigma_\eta^2)$, and $v_{it} \sim iid(0, \sigma_v^2)$. These are not crucial for the main consistency arguments, but they simplify some derivations and motivate standard GLS formulas.

A simple approach to estimate $\boldsymbol{\beta}$ is to ignore the error structure. This is, define $\boldsymbol{u} \equiv \boldsymbol{\eta} + \boldsymbol{v}$, and estimate $\boldsymbol{\beta}$ by OLS:

$$\hat{\boldsymbol{\beta}}_{OLS} = (X'X)^{-1}X'\boldsymbol{y}. \tag{5}$$

The properties of $\hat{\boldsymbol{\beta}}_{OLS}$ depend on whether $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] = 0$, because under strict exogeneity we have $\mathbb{E}[\boldsymbol{x}_{it}v_{it}] = 0$, and hence $\mathbb{E}[\boldsymbol{x}_{it}u_{it}] = \mathbb{E}[\boldsymbol{x}_{it}\eta_i]$. If $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] = 0$ (random effects), $\hat{\boldsymbol{\beta}}_{OLS}$ is consistent as $N \to \infty$ or $T \to \infty$ or both. However, it is efficient only if $\sigma^2_\eta = 0$, i.e. if $\eta_i = 0\ \forall i$. If $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] \neq 0$ (fixed effects), $\hat{\boldsymbol{\beta}}_{OLS}$ is inconsistent as $N \to \infty$ or $T \to \infty$ or both. Note that cross-section results are also inconsistent, and panel data helps in constructing a consistent alternative.

In our example, this would imply redefining Equation (1) as:

$$\ln C_j = \beta_0 + \beta_1 \ln P_j + \beta_2 \ln Y_j + u_j, \tag{6}$$

where subindex $j$ is used to emphasize that observations $it$ are considered to be independent from each other. OLS estimation of this equation delivers the following result (standard errors in parentheses):

| | OLS | |
|---|---|---|
| ln Prices ($\beta_1$) | -0.083 | (0.015) |
| ln Income ($\beta_2$) | -0.032 | (0.006) |

Note that this estimation may suffer from potential problems. Some individuals may have a higher propensity to smoke than others, everything else equal. This higher propensity could be correlated with observable factors. For example, maybe individuals whose parents used to smoke are more likely to smoke, and that may correlate with income. Likewise, individuals at different ages may have taste for different brands of tobacco that may have different pricing. In both cases, these omitted variables introduce some spurious correlation between regressors and dependent variable that can induce obtaining biased estimates of cigarette demand.

### *A. The Fixed Effects Model. Within Groups Estimation*

We first assume correlated (fixed) effects: $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] \neq 0$. As already noted, OLS estimation of such model delivers inconsistent estimates. As an alternative, consider the transformation of the model in deviations from individual means, $\tilde{y}_{it} \equiv y_{it} - \bar{y}_i$, where $\bar{y}_i \equiv T^{-1}\sum_{t=1}^{T} y_{it}$:

$$\tilde{y}_{it} = (\boldsymbol{x}_{it} - \bar{\boldsymbol{x}}_i)'\boldsymbol{\beta} + (\eta_i - \bar{\eta}_i) + (v_{it} - \bar{v}_i) = \tilde{\boldsymbol{x}}'_{it}\boldsymbol{\beta} + \tilde{v}_{it}. \tag{7}$$

Note that the transformation eliminates the individual effect, as $\bar{\eta}_i = \eta_i$. As a result, and given our earlier assumptions, $\mathbb{E}[\tilde{\boldsymbol{x}}_{it}\tilde{v}_{it}] = 0$, and OLS on the transformed model delivers consistent estimates. Such estimator is called the *within*

*groups estimator*, and can be written as:

$$\hat{\boldsymbol{\beta}}_{WG} = \left(\tilde{X}'\tilde{X}\right)^{-1} \tilde{X}'\tilde{y}. \tag{8}$$

This estimator is consistent regardless of whether $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] \neq 0$ or $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] = 0$ if strict exogeneity, $\mathbb{E}[\boldsymbol{x}_{it}v_{is}] = 0\ \forall s, t$, is satisfied. Strict exogeneity is crucial for consistency when $N \to \infty$ and $T$ is fixed. The reason is that the within transformation uses time averages that depend on the entire history of the regressors and shocks for individual $i$:

$$\tilde{\boldsymbol{x}}_{it} = \boldsymbol{x}_{it} - \frac{1}{T}\sum_{s=1}^{T} \boldsymbol{x}_{is}, \qquad \tilde{v}_{it} = v_{it} - \frac{1}{T}\sum_{s=1}^{T} v_{is}.$$

Therefore, $\tilde{\boldsymbol{x}}_{it}$ is a function of $(\boldsymbol{x}_{i1}, \ldots, \boldsymbol{x}_{iT})$ and $\tilde{v}_{it}$ is a function of $(v_{i1}, \ldots, v_{iT})$. To ensure $\mathbb{E}[\tilde{\boldsymbol{x}}_{it}\tilde{v}_{it}] = 0$, we generally need

$$\mathbb{E}[\boldsymbol{x}_{it}v_{is}] = 0 \quad \forall s, t,$$

unless we take $T \to \infty$ (in which case the influence of the time average becomes negligible). This is one of the motivations for dynamic panel methods, which are designed to relax strict exogeneity.

The consistency of the within groups estimator regardless of whether we are in a correlated or uncorrelated effects situation (under strict exogeneity) is an important advantage relative to pooled OLS and random-effects GLS. However, this advantage comes at a cost. First, the within estimator is generally *not efficient* when $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] = 0$ and $N \to \infty$ with $T$ fixed: random-effects GLS (when correctly specified) exploits the error-components covariance structure and is more efficient. Second, even under fixed effects, the within transformation induces correlation in the transformed errors because $\tilde{v}_{it} = v_{it} - \bar{v}_i$ shares $\bar{v}_i$ across all $t$ for a given $i$, so naive homoskedastic/serially independent formulas are inappropriate (one typically uses cluster-robust standard errors at the individual level). Finally, the within estimator does not identify coefficients on *time-invariant* regressors (and it identifies poorly nearly time-invariant regressors). Identification comes from *within-unit variation*, i.e. from "switchers" in the corresponding regressor.

In our example, within groups estimates deliver the following results:

| | OLS | | WG | |
|---|---|---|---|---|
| ln Prices ($\beta_1$) | -0.083 | (0.015) | -0.292 | (0.023) |
| ln Income ($\beta_2$) | -0.032 | (0.006) | 0.107 | (0.019) |

The within groups estimator can also be obtained by including a set of $N$ individual dummy variables in the regression:

$$y_{it} = \boldsymbol{x}_{it}'\boldsymbol{\beta} + \eta_1 D_{1i} + ... + \eta_N D_{Ni} + v_{it}, \tag{9}$$

where $D_{hi} = \mathbb{1}\{h = i\}$ (e.g. $D_{1i}$ takes a value of 1 for the observations on individual 1 and 0 for all other observations). OLS estimation of this model gives estimates that are numerically equivalent to within groups. For this reason, the within groups estimator is also known as *least squares dummy variables* estimator.

An alternative transformation of the model that also eliminates the individual effects is first-differencing:

$$\Delta y_{it} = \Delta \boldsymbol{x}_{it}'\boldsymbol{\beta} + \Delta v_{it}, \tag{10}$$

where $\Delta y_{it} = y_{it} - y_{it-1}$. The fact that this transformation eliminates individual effects is evident from that they are time-invariant ($\Delta\eta_i = \eta_i - \eta_i = 0$). Therefore, OLS estimation on the differenced model is consistent. An advantage of *first-differenced least squares* is that it only requires $\mathbb{E}[\Delta\boldsymbol{x}_{it}\Delta v_{it}] = 0$ for consistency, which is implied by but weaker than strict exogeneity. However, under the classical assumptions, within groups is more efficient. First-differenced least squares, nonetheless, is more efficient if $v_{it}$ is a random walk.

In our example, Least Squares Dummy Variables (LSDV) and and first-differenced least squares (FDLS) are presented in the following table:

### *B. The Random Effects Model. Error Components*

Now we turn into the assumption of uncorrelated or random effects: $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] = 0$. In this case, OLS is consistent, but not efficient. The inefficiency is provided by the serial correlation introduced by $\eta_i$:

$$\mathbb{E}[u_{it}u_{is}] = \mathbb{E}[(\eta_i + v_{it})(\eta_i + v_{is})] = \mathbb{E}[\eta_i^2] = \sigma_\eta^2. \tag{11}$$

Likewise, the variance of the unobservables is:

$$\mathbb{E}[u_{it}^2] = \mathbb{E}[\eta_i^2] + \mathbb{E}[v_{it}^2] = \sigma_\eta^2 + \sigma_v^2. \tag{12}$$

Therefore, the variance-covariance matrix of the unobservables is a block-diagonal matrix formed by elements:

$$\mathbb{E}[\boldsymbol{u}_i\boldsymbol{u}_i'] = \begin{pmatrix} \sigma_\eta^2 + \sigma_v^2 & \sigma_\eta^2 & \dots & \sigma_\eta^2 \\ \sigma_\eta^2 & \sigma_\eta^2 + \sigma_v^2 & \dots & \sigma_\eta^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_\eta^2 & \sigma_\eta^2 & \dots & \sigma_\eta^2 + \sigma_v^2 \end{pmatrix} = \Omega_i, \tag{13}$$

| | OLS | WG | LSDV | FDLS |
|---|---|---|---|---|
| ln Prices ($\beta_1$) | -0.083 | -0.292 | -0.292 | -0.413 |
| | (0.015) | (0.023) | (0.023) | (0.035) |
| ln Income ($\beta_2$) | -0.032 | 0.107 | 0.107 | 0.178 |
| | (0.006) | (0.019) | (0.019) | (0.055) |
| Individual 1 ($\beta_0 + \eta_1$) | | | 2.804 | |
| | | | (0.288) | |
| Individual 2 ($\beta_0 + \eta_2$) | | | 3.455 | |
| | | | (0.398) | |
| Individual 3 ($\beta_0 + \eta_3$) | | | 2.891 | |
| | | | (0.416) | |
| Individual 4 ($\beta_0 + \eta_4$) | | | 2.908 | |
| | | | (0.384) | |
| Individual 5 ($\beta_0 + \eta_5$) | | | 3.490 | |
| | | | (0.433) | |
| Individual 6 ($\beta_0 + \eta_6$) | | | 2.092 | |
| | | | (0.325) | |
| Individual 7 ($\beta_0 + \eta_7$) | | | 1.769 | |
| | | | (0.393) | |
| ... | | | ... | |

so that:

$$\mathbb{E}[\boldsymbol{u}\boldsymbol{u}'] = \begin{pmatrix} \Omega_1 & 0 & \dots & 0 \\ 0 & \Omega_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \Omega_N \end{pmatrix} = \Omega. \tag{14}$$

Under the classical assumptions and random effects, a GLS estimator that incorporates this covariance structure, as noted by Balestra and Nerlove (1966), is consistent and efficient:

$$\hat{\boldsymbol{\beta}}_{GLS} = (X'\Omega^{-1}X)^{-1}X'\Omega^{-1}\boldsymbol{y}. \tag{15}$$

This estimator is unfeasible, as $\sigma_\eta^2$ and $\sigma_v^2$ are unknown.

Consistency crucially relies on $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] = 0$. It is convenient to rewrite GLS as OLS on the *quasi-demeaned* (theta-differenced) model:

$$y_{it}^* = \boldsymbol{x}_{it}^{*\prime}\boldsymbol{\beta} + u_{it}^*, \qquad y_{it}^* \equiv y_{it} - \theta\,\bar{y}_i, \qquad \boldsymbol{x}_{it}^* \equiv \boldsymbol{x}_{it} - \theta\,\bar{\boldsymbol{x}}_i, \tag{16}$$

where

$$\theta = 1 - \sqrt{\frac{\sigma_v^2}{\sigma_v^2 + T\sigma_\eta^2}}. \tag{17}$$

Note that this transformation does not eliminate $\eta_i$ in general, so consistency of

random-effects GLS requires $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] = 0$.This representation is useful to understand two special cases:

- If $\sigma_\eta^2 = 0$, then $\theta = 0$ and $(y_{it}^*, \boldsymbol{x}_{it}^*) = (y_{it}, \boldsymbol{x}_{it})$, so GLS reduces to pooled OLS (and pooled OLS is efficient).
- If $T \to \infty$, then $\theta \to 1$ and $(y_{it}^*, \boldsymbol{x}_{it}^*) \to (y_{it} - \bar{y}_i, \boldsymbol{x}_{it} - \bar{\boldsymbol{x}}_i)$, so GLS approaches the within estimator (and the within estimator becomes efficient).

A feasible GLS estimator is obtained by recovering consistent estimates of $\sigma_\eta^2$ and $\sigma_v^2$. A consistent estimator of $\sigma_v^2$ is provided by the within groups residuals:

$$\hat{\tilde{v}}_{it} \equiv \tilde{y}_{it} - \tilde{\boldsymbol{x}}_{it}'\hat{\boldsymbol{\beta}}_{WG} \quad \Rightarrow \quad \hat{\sigma}_v^2 = \frac{\hat{\tilde{\boldsymbol{v}}}'\hat{\tilde{\boldsymbol{v}}}}{N(T-1)-K}. \tag{18}$$

To obtain an estimate of $\sigma_\eta^2$, we need to recover the residuals of a *between groups* estimation, which is the OLS estimation on the cross-section of individual averages:

$$\bar{y}_i = \bar{\boldsymbol{x}}_i'\boldsymbol{\beta} + \bar{\eta}_i + \bar{v}_i, \quad i = 1, ..., N. \tag{19}$$

In particular, $\sigma_\eta^2$ is recovered from the residuals of that regression, using the estimated $\hat{\sigma}_v^2$:

$$\hat{\bar{u}}_i \equiv \bar{y}_i - \bar{\boldsymbol{x}}_i'\hat{\boldsymbol{\beta}}_{BG} \quad \Rightarrow \quad \hat{\sigma}_{\bar{u}}^2 = \widehat{\left(\sigma_\eta^2 + \frac{1}{T}\sigma_v^2\right)} = \frac{\hat{\bar{\boldsymbol{u}}}'\hat{\bar{\boldsymbol{u}}}}{N-K} \quad \Rightarrow \quad \hat{\sigma}_\eta^2 = \hat{\sigma}_{\bar{u}}^2 - \frac{1}{T}\hat{\sigma}_v^2. \tag{20}$$

In our example, Feasible GLS results are in the following table:

| | OLS | WG | FGLS |
|---|---|---|---|
| ln Prices ($\beta_1$) | -0.083 | -0.292 | -0.122 |
| | (0.015) | (0.023) | (0.014) |
| ln Income ($\beta_2$) | -0.032 | 0.107 | -0.012 |
| | (0.006) | (0.019) | (0.004) |

### *C. Testing for Correlated Individual Effects*

Given the efficiency advantage of feasible GLS compared to within groups when the random effects assumption is satisfied, but its inconsistency otherwise, it is useful to test whether we are in a random or fixed effects situation. As $\hat{\boldsymbol{\beta}}_{WG}$ is consistent under the two situations, but $\hat{\boldsymbol{\beta}}_{FGLS}$ is only consistent in the random effects case, we can test wether they are similar.

This comparison is done by the Hausman test (Hausman, 1978):

$$h \equiv \hat{\boldsymbol{q}}'[avar(\hat{\boldsymbol{q}})]^{-1}\hat{\boldsymbol{q}} \overset{a}{\sim} \chi^2(K) \tag{21}$$

under the null hypothesis of $\mathbb{E}[\boldsymbol{x}_{it}\eta_i] = 0$, where:

$$\hat{\boldsymbol{q}} = \hat{\boldsymbol{\beta}}_{WG} - \hat{\boldsymbol{\beta}}_{FGLS}, \quad \text{and} \quad avar(\hat{\boldsymbol{q}}) = avar\left(\hat{\boldsymbol{\beta}}_{WG}\right) - avar\left(\hat{\boldsymbol{\beta}}_{FGLS}\right). \tag{22}$$

The test requires the classical assumptions to be satisfied, in order to ensure that feasible GLS is more efficient than within groups.

In our example:

| | Statistic | P-value |
|---|---|---|
| Hausman test | 24.661 | 0.000 |

## III. Dynamic Models

We now include feedback from past shocks into future outcomes, relaxing strict exogeneity. In our cigarette consumption model, one may want to introduce past consumption of cigarettes to capture the effect of tobacco addiction (the more you smoke one period, the more you want to smoke next period). We focus on first order autoregressive models, but the results can be generalized to other forms of persistence, and to the inclusion of regressors.

### *A. Autoregressive Models with Individual Effects*

We consider the following model:

$$y_{it} = \alpha y_{it-1} + \eta_i + v_{it} \quad |\alpha| < 1. \tag{23}$$

We assume that we observe $y_{i0}$, and, hence, our sample includes observations for $i = 1, ..., N$, and $t = 1, ..., T$. We keep the error components and serially uncorrelated shocks assumptions (and potentially the homoskedasticity/iid), but we relax strict exogeneity. Instead, we assume a much weaker condition, namely predetermined initial condition: $\mathbb{E}[y_{i0}v_{it}] = 0$ for $t = 1, ..., T$. Note that we are in a fixed effects situation as $\mathbb{E}[y_{it-1}\eta_i] \neq 0$ by construction. Likewise, $\mathbb{E}[y_{it-1}v_{it-1}] \neq 0$.

Even if $\mathbb{E}[y_{it-1}v_{it}] = 0$, OLS is biased. In particular:

$$\operatorname*{plim}_{N\to\infty} \hat{\alpha}_{OLS} > \alpha, \tag{24}$$

because $\mathbb{E}[y_{it-1}\eta_i] = \sigma_\eta^2 \left(\frac{1-\alpha^{t-1}}{1-\alpha}\right) + \alpha^{t-1}\,\mathbb{E}[y_{i0}\eta_i] > 0$. Likewise, within groups is biased because $\mathbb{E}[\tilde{y}_{it-1}\tilde{v}_{it}] \neq 0$ (Nickell Bias). In particular:

$$\operatorname*{plim}_{N\to\infty} \hat{\alpha}_{WG} < \alpha, \tag{25}$$

because $\mathbb{E}[\tilde{y}_{it-1}\tilde{v}_{it}] = -A\sigma_v^2 < 0$, with $\left(A = \frac{(1-\alpha)\left(1+T\left(1-\alpha^{t-1}-\alpha^{T-1-t}\right)\right)+\alpha T\left(1-\alpha^{T-1}\right)}{T^2(1-\alpha)^2}\right)$. The key problem is that after the within transformation, $\tilde{y}_{it-1}$ contains $\bar{v}_i$, which in turn contains $v_{it}$. Therefore $\tilde{y}_{it-1}$ is mechanically correlated with the transformed error $\tilde{v}_{it}$, generating the so-called *Nickell bias*. Note that the Nickell bias vanishes as $T \to \infty$, but, in practice, the bias is not small even with reasonably long micro panels such as $T = 15$. Given the sign of the biases, OLS and within groups give interesting bounds, and estimators that give $\hat{\alpha} >> \alpha_{OLS}$ or $\hat{\alpha} << \hat{\alpha}_{WG}$ should be seen with suspicion.

A seminal approach to correct these biases was proposed by Anderson and Hsiao (1981, 1982). Consider the model in first differences:

$$\Delta y_{it} = \alpha \Delta y_{it-1} + \Delta v_{it}. \tag{26}$$

OLS in first differences is inconsistent, because $\mathbb{E}[\Delta y_{it-1}\Delta v_{it}] = -\sigma_v^2 < 0$. However, if the serially uncorrelated shocks assumption holds, $\mathbb{E}[y_{it-1}v_{it}] = 0$ and $y_{it-2}$ or $\Delta y_{it-2}$ are valid instruments for $\Delta y_{it-1}$. In particular, they satisfy relevance, as $\mathbb{E}[\Delta y_{it-2}\Delta y_{it-1}] \neq 0$ and $\mathbb{E}[y_{it-2}\Delta y_{it-1}] \neq 0$, and orthogonality, as $\mathbb{E}[\Delta y_{it-2}\Delta v_{it}] = \mathbb{E}[y_{it-2}\Delta v_{it}] = 0$. Therefore, the Anderson-Hisao estimator is:

$$\hat{\alpha}_{AH} = \left(\widehat{\Delta \boldsymbol{y}'_{-1}}\widehat{\Delta \boldsymbol{y}_{-1}}\right)^{-1} \widehat{\Delta \boldsymbol{y}'_{-1}}\Delta \boldsymbol{y}, \tag{27}$$

where

$$\widehat{\Delta \boldsymbol{y}_{-1}} = Z\left(Z'Z\right)^{-1} Z'\Delta \boldsymbol{y}_{-1}, \tag{28}$$

and $Z$ is either $\boldsymbol{y}_{-2}$ or $\Delta \boldsymbol{y}_{-2}$. A minimum of three periods ($T = 2$ plus $y_{i0}$) is needed to implement it. This approach is only efficient if $T = 2$ (otherwise, additional instruments are available, as discussed below).

In our example, we redefine the model to be an AR(1) process (for now without regressors):

$$n_{it} = \alpha n_{it-1} + \eta_i + v_{it} \tag{29}$$

The Anderson-Hsiao results (together with OLS and WG) are:

### *B. A small digression: quick review of Generalized Method of Moments (GMM)*

GMM provides a unified way to estimate parameters by choosing them so that sample analogues of population *orthogonality conditions* are as close to zero as

| | OLS | WG | Anderson-Hsiao |
|---|---|---|---|
| Lagged cons. ($\ln C_{it-1}$) | 0.982 | 0.884 | 1.195 |
| | (0.003) | (0.061) | (0.190) |

possible. Conceptually, it is useful to organize GMM in three pieces: (i) the setup (moments), (ii) the estimation (just-identified vs overidentified), and (iii) asymptotic properties (consistency and variance).

We are interested in a parameter vector $\boldsymbol{\beta}$ that is defined by the set of moments (or orthogonality conditions) given by:

$$\mathbb{E}[\psi(\boldsymbol{x};\boldsymbol{\beta})] = \mathbf{0}, \tag{30}$$

where $\boldsymbol{x}$ is a (vector) random variable, $\boldsymbol{\beta}$ is the parameter vector, and $\psi(\ )$ is a vector function such that $\dim(\psi) \geq \dim(\boldsymbol{\beta})$. Therefore, the problem specifies $\dim(\psi)$ moment conditions.

For example, consider a regression model. The parameter vector, $\boldsymbol{\beta}$, is such that:

$$\mathbb{E}[\boldsymbol{z}u] = \mathbf{0}, \tag{31}$$

where:

$$u \equiv y - f(\boldsymbol{x};\boldsymbol{\beta}), \quad \text{and} \quad \boldsymbol{z} \equiv g(\boldsymbol{x}), \tag{32}$$

with $\dim(\boldsymbol{z}) \geq \dim(\boldsymbol{\beta})$.

We have a sample of $N$ observations $\{\boldsymbol{x}_i\}_{i=1}^N$. The estimation is based on the sample analog of (30):

$$\boldsymbol{b}_N(\boldsymbol{\theta}) = \frac{1}{N}\sum_{\mathrm{i=1}}^{\mathrm{N}} \psi(\boldsymbol{x}_i;\boldsymbol{\beta}). \tag{33}$$

The GMM estimator is given by the value of $\boldsymbol{\beta}$ that minimizes the quadratic distance of $b_N(\boldsymbol{\beta})$ from zero:

$$\hat{\boldsymbol{\beta}}_{GMM} = \arg\min_{\beta\in\Theta} \boldsymbol{b}_N(\boldsymbol{\beta})' W_N \boldsymbol{b}_N(\boldsymbol{\beta}), \tag{34}$$

where $W_N$ is a squared semi-positive definite weighting matrix that satisfies the rank condition $\mathrm{rank}(W_N) \geq \dim(\boldsymbol{\beta})$. Note that, if the problem is just-identified, this is when $\dim(\psi) = \dim(\boldsymbol{\beta})$, the weighting matrix becomes irrelevant, and the GMM estimator satisfies:

$$\boldsymbol{b}_N(\hat{\boldsymbol{\beta}}_{GMM}) = \mathbf{0}. \tag{35}$$

Building on the previous example, consider the linear regression model, i.e.

$f(\boldsymbol{x};\boldsymbol{\beta}) = \boldsymbol{x}'\boldsymbol{\beta}$. Then:

$$\boldsymbol{b}_N(\boldsymbol{\beta}) = N^{-1}\sum_{\mathrm{i=1}}^{\mathrm{N}} \boldsymbol{z}_i(y_i - \boldsymbol{x}_i'\boldsymbol{\beta}) = N^{-1}Z'(\boldsymbol{y} - X\beta), \tag{36}$$

and $\hat{\boldsymbol{\beta}}_{GMM}$ satisfies:

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{GMM} &= \arg\min_{\beta} N^{-2}(\boldsymbol{y} - X\beta)'ZW_NZ'(\boldsymbol{y} - X\beta) \\ &= (X'ZW_NZ'X)^{-1}X'ZW_NZ'\boldsymbol{y},\end{aligned} \tag{37}$$

which is a familiar expression, as it equals 2SLS if $W_N = (Z'Z)^{-1}$.

Under some general conditions, $\hat{\boldsymbol{\beta}}_{GMM}$ is a consistent estimator of $\boldsymbol{\beta}$. Additionally, it is asymptotically normal, with the following variance:

$$avar(\hat{\boldsymbol{\beta}}_{GMM}) = (D'WD)^{-1}D'WS_0WD(D'WD)^{-1}, \tag{38}$$

where $D \equiv \mathrm{plim}_{N\to\infty}\frac{\partial \boldsymbol{b}_N(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'}$, $W \equiv \mathrm{plim}_{N\to\infty} W_N$, and $S_0 \equiv \frac{1}{N}\sum_{\mathrm{i=1}}^{\mathrm{N}} \mathbb{E}[\boldsymbol{z}_i u_i u_i' \boldsymbol{z}_i']$.

Even though any semi-positive definite weighting matrix that satisfies the rank condition provides a consistent estimate of $\boldsymbol{\beta}$, not all of them form an efficient estimator. Efficiency is achieved with any $W_N$ that implies $W_0 = \kappa S_0^{-1}$, for a positive $\kappa$. This includes $W_N = S_0^{-1}$ (unfeasible), but also $W_N = \hat{S}_N^{-1}$, where $\hat{S}_N$ is any consistent estimator of $S_0$. In practice, the Optimal GMM estimator is implemented in two steps:

1) Obtain $\hat{\boldsymbol{\beta}}_{GMM}(W_N^0)$ for an initial guess $W_N^0$.
2) Re-estimate using $W_{opt} = (\sum_{\mathrm{i=1}}^{\mathrm{N}} \psi(\boldsymbol{x}_i;\hat{\boldsymbol{\beta}}_{GMM}(W_N^0))\psi(\boldsymbol{x}_i;\hat{\boldsymbol{\beta}}_{GMM}(W_N^0))')^{-1}$ as the new weighting matrix.

### *C. Difference GMM Estimation*

In a very influential paper, Arellano and Bond (1991), proposed a GMM estimation that uses all available exogenous variation in the estimation. In particular, they use the fact that for a given period $t$, not only $y_{it-2}$ satisfy relevance and orthogonality conditions, but also $y_{it-3}, ..., y_{i0}$ do so. Therefore, the Arellano-Bond is defined by the following $(T-1)T/2$ orthogonality conditions:

$$\mathbb{E}[\Delta v_{i2}y_{i0}] = 0, \quad \mathbb{E}\left[\Delta v_{i3}\begin{pmatrix} y_{i0} \\ y_{i1}\end{pmatrix}\right] = \mathbf{0}, \quad \ldots, \quad \mathbb{E}\left[\Delta v_{iT}\begin{pmatrix} y_{i0} \\ y_{i1} \\ y_{i2} \\ \vdots \\ y_{iT-2}\end{pmatrix}\right] = \mathbf{0}. \tag{39}$$

It is convenient to write these moment conditions as $\mathbb{E}[Z_i'\Delta\boldsymbol{v}_i] = 0$, where:

$$Z_i = \begin{pmatrix} y_{i0} & 0 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & y_{i0} & y_{i1} & 0 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & 0 & \dots & y_{i0} & y_{i1} & \dots & y_{iT-2} \end{pmatrix} \text{ and } \Delta\boldsymbol{v}_i = \begin{pmatrix} \Delta v_{i2} \\ \Delta v_{i3} \\ \vdots \\ \Delta v_{iT} \end{pmatrix}. \tag{40}$$

The sample analogue is:

$$\boldsymbol{b}_N(\alpha) = \frac{1}{N}\sum_{\mathrm{i=1}}^{\mathrm{N}} Z_i'\Delta\boldsymbol{v}_i(\alpha), \tag{41}$$

and the Arellano-Bond estimator is:

$$\begin{aligned} \hat{\alpha}_{GMM} &= \arg\min_{\alpha} \left(\frac{1}{N}\sum_{\mathrm{i=1}}^{\mathrm{N}} \Delta\boldsymbol{v}_i'(\alpha)Z_i\right) W_N \left(\frac{1}{N}\sum_{\mathrm{i=1}}^{\mathrm{N}} Z_i'\Delta\boldsymbol{v}_i(\alpha)\right) \\ &= (\Delta\boldsymbol{y}_{-1}'ZW_NZ'\Delta\boldsymbol{y}_{-1})^{-1}\Delta\boldsymbol{y}_{-1}'ZW_NZ'\Delta\boldsymbol{y}. \end{aligned} \tag{42}$$

In order to obtain efficient estimates, the optimal weighting matrix should be used. The unfeasible optimal weighting matrix is:

$$W_N = \left(\frac{1}{N}\sum_{\mathrm{i=1}}^{\mathrm{N}} \mathbb{E}[Z_i'\Delta\boldsymbol{v}_i\Delta\boldsymbol{v}_i'Z_i]\right)^{-1}, \tag{43}$$

and a feasible and asymptotically equivalent alternative is obtained in two-steps as:

$$W_N = \left(\frac{1}{N}\sum_{\mathrm{i=1}}^{\mathrm{N}} [Z_i'\widehat{\Delta\boldsymbol{v}_i(\hat{\alpha})}\widehat{\Delta\boldsymbol{v}_i'(\hat{\alpha})}Z_i]\right)^{-1}. \tag{44}$$

Windmeijer (2005) proposes a finite sample correction of the variance that takes into account that $\alpha$ is estimated.

A common one-step (and first-step) weighting matrix uses the implied structure of $\mathbb{E}[\Delta\boldsymbol{v}_i\Delta\boldsymbol{v}_i']$ and sets:

$$W = (Z'\ \mathbb{E}[\Delta\boldsymbol{v}_i\Delta\boldsymbol{v}_i']\ Z)^{-1}, \tag{45}$$

where:

$$\mathbb{E}[\Delta\boldsymbol{v}_i\Delta\boldsymbol{v}_i'] = \sigma_v^2 \begin{pmatrix} 2 & -1 & 0 & 0 & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ 0 & 0 & 0 & 0 & \dots & 2 \end{pmatrix}. \tag{46}$$

Note that any positive scale factor (e.g. $\sigma_v^2$) is irrelevant for the minimization problem, so, taking $\sigma_v^2$ out of the weighting matrix, $W$ can be directly computed from the data.

There are two main shortcomings with the Arellano–Bond approach. First, *weak instruments*. When $\alpha \to 1$, lagged levels become weak instruments for lagged differences, so relevance deteriorates. While the instruments remain valid as long as $|\alpha| < 1$, small sample properties can be poor. Monte Carlo evidence suggests that when $\alpha$ is large (e.g. above 0.8), very large samples may be needed for good performance. This motivates alternatives such as System GMM. Second, *instrument proliferation / overfitting*. The number of available moment conditions grows on the order of $(T-1)T/2$, which can be large when $T$ is not small relative to $N$. In this case, GMM may overfit endogenous variables and weaken overidentification tests. A common practice is to restrict (collapse) the instrument set and check robustness across instrument choices.

The GMM results in our AR(1) example (including one-step, two-step, and two-step with small sample correction) are:

| | Coefficient | Standard Error |
|---|---|---|
| Least Squares (OLS) | 0.982 | (0.003) |
| Within Groups (WG) | 0.884 | (0.061) |
| Anderson-Hsiao | 1.195 | (0.190) |
| One-step GMM | 1.023 | (0.104) |
| Two-step GMM | 0.994 | (0.040) |
| Two-step GMM small sample | 0.994 | (0.121) |

The extension of this approach to models that include regressors is straightforward. Consider the following model:

$$y_{it} = \alpha y_{it-1} + \boldsymbol{x}_{it}'\boldsymbol{\beta} + \eta_i + v_{it} \quad |\alpha| < 1. \tag{47}$$

We maintain the previous assumptions: error components, serially uncorrelated shocks, and predetermined initial conditions. Therefore, the moment conditions in Equation (39) are still valid. Different assumptions regarding $\boldsymbol{x}_{it}$ will add to them different additional orthogonality conditions. Specifically, $\boldsymbol{x}_{it}$ can be correlated or uncorrelated with $\boldsymbol{\eta_i}$, and $\boldsymbol{x}_{it}$ can be endogenous, predetermined, or strictly exogenous with respect to $v_{it}$. In either of this cases, different instruments will be used in the new orthogonality conditions. For instance, if assumptions are analogous to those for $y_{it-1}$, we may use $\boldsymbol{x}_{it-1}$ and previous lags as instruments for $\boldsymbol{x}_{it}$. In that case, the matrix of instruments would be expanded as follows:

$$Z_i = \begin{pmatrix} y_{i0} & \boldsymbol{x}_{i0}' & \boldsymbol{x}_{i1}' & \dots & 0 & \dots & 0 & \boldsymbol{0}' & \dots & \boldsymbol{0}' \\ \vdots & \vdots & \vdots & \ddots & \vdots & \dots & \vdots & \vdots & \vdots & \vdots \\ 0 & \boldsymbol{0}' & \boldsymbol{0}' & \dots & y_{i0} & \dots & y_{iT-2} & \boldsymbol{x}_{i0}' & \dots & \boldsymbol{x}_{iT-1}' \end{pmatrix}. \tag{48}$$

In our example, we rewrite the log-cigarette consumption equations as follows:

$$\ln C_{it} = \alpha \ln C_{it-1} + \beta_1 \ln P_{it} + \beta_2 \ln Y_{it} + \eta_i + v_{it}. \tag{49}$$

Results are:

| | OLS | WG | GMM |
|---|---|---|---|
| Lagged dep ($\alpha$) | 0.947 | 0.528 | 0.495 |
| | (0.011) | (0.064) | (0.127) |
| ln Prices ($\beta_1$) | 0.010 | -0.501 | -0.607 |
| | (0.006) | (0.098) | (0.143) |
| ln Income ($\beta_2$) | 0.049 | 0.369 | 0.338 |
| | (0.011) | (0.044) | (0.051) |

### *D. System GMM Estimation*

The System GMM estimator, proposed by Arellano and Bover (1995) uses the assumption $\mathbb{E}[y_{i0}|\eta_i] = \frac{\eta_i}{1-\alpha}$, which provides additional moment conditions. In particular, this implies that $\mathbb{E}[\Delta y_{it}\eta_i] = 0$ for any $t$, or, equivalently:

$$\mathbb{E}[\Delta y_{iT-s} u_{iT}] = 0, \quad u_{iT} \equiv \eta_i + v_{iT}, \tag{50}$$

for $s = 1, ..., T-1$. Therefore, we rewrite the moment conditions as $\mathbb{E}[(Z^*)' u_i^*] = 0$, with:

$$Z_i^* = \begin{pmatrix} Z_i & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0}' & \Delta y_{i1} & \dots & \Delta y_{iT-1} \end{pmatrix} \quad \text{and} \quad \boldsymbol{u}_i^* = \begin{pmatrix} \Delta \boldsymbol{v}_i \\ \eta_i + v_{iT} \end{pmatrix}, \tag{51}$$

and the System GMM estimator is:

$$\hat{\alpha}_{Sys-GMM} = \left(X^{*\prime} Z^* W_N Z^{*\prime} X^*\right)^{-1} X^* Z^* W_N Z^{*\prime} \boldsymbol{y}^*, \tag{52}$$

where:

$$X_i^* = \begin{pmatrix} \Delta \boldsymbol{y}_{-1i} \\ y_{iT-1} \end{pmatrix} \quad \text{and} \quad \boldsymbol{y}_i^* = \begin{pmatrix} \Delta \boldsymbol{y}_i \\ y_{iT} \end{pmatrix}. \tag{53}$$

This estimator is more efficient that differenced GMM, as it uses additional moment conditions. It also reduces the small sample bias, especially when $\alpha \to 1$.

Adding System-GMM results to the previous AR(1) results:

### *E. Specification Tests*

There are several relevant aspects for the validity of the estimation that can be tested formally. The null hypothesis that the orthogonality conditions are satisfied (i.e. moments are equal to zero) can be tested in general, as the estimation

| | Coefficient | Standard Error |
|---|---|---|
| Least Squares (OLS) | 0.982 | (0.003) |
| Within Groups (WG) | 0.884 | (0.061) |
| Anderson-Hsiao | 1.195 | (0.190) |
| One-step GMM | 1.023 | (0.104) |
| Two-step GMM | 0.994 | (0.040) |
| Two-step GMM small sample | 0.994 | (0.121) |
| One-step System-GMM | 0.926 | (0.023) |
| Two-step System-GMM small | 0.911 | (0.032) |

problem is typically overidentified (if $T > 2$). The standard Sargan/Hansen overidentifying restrictions test is applicable (Sargan, 1958; Hansen, 1982). The null hypothesis is that the orthogonality conditions are satisfied (i.e. the population moments are zero). When the model is overidentified, the GMM $J$-test statistic is:

$$J = N\,\hat{g}(\hat{\beta})'\,\hat{S}^{-1}\,\hat{g}(\hat{\beta}), \qquad \hat{g}(\hat{\beta}) \equiv \frac{1}{N}\sum_{i=1}^{N} Z_i'\hat{u}_i. \tag{54}$$

Under the null that all instruments are valid, $J \overset{a}{\sim} \chi^2(L-K)$, where $L$ is the number of moment conditions and $K$ the number of parameters.

Some of the assumptions that we impose in dynamic panel models are stronger than others. For example, System-GMM relies on additional orthogonality conditions relative to Difference-GMM. If these additional moment conditions are valid, they improve efficiency. If they are not valid, the estimator becomes inconsistent. Therefore, it is useful to test whether the additional orthogonality conditions implied by stronger assumptions are supported by the data.

There are two closely related ways to do this. The first one is based on the *Incremental (difference) Sargan/Hansen test.*. Let $J^{Diff-GMM}$ and $J^{Sys-GMM}$ denote the corresponding overidentifying restrictions statistics for difference and system GMM. If the additional system GMM moments are valid, the difference in the two $J$-statistics should be small (i.e., the new moment conditions are also close to zero). Formally, the incremental test statistic is:

$$\Delta J = J^{Sys-GMM} - J^{Diff-GMM} \overset{a}{\sim} \chi^2\big(L^{Sys-GMM} - L^{Diff-GMM}\big), \tag{55}$$

where $L$ denotes the number of moment conditions in each model. Under the null hypothesis that the additional moment conditions are valid, $\Delta J$ follows a chi-squared distribution with degrees of freedom equal to the number of additional moments.

The second option is to run a *Hausman test based on coefficient differences.* An alternative way of testing the same idea is to compare the coefficient estimates

from the two models. If the stronger assumptions are valid, both estimators are consistent, but the more efficient one (e.g. System-GMM) should dominate in terms of precision. Define $\hat{\boldsymbol{q}} = \hat{\hat{\boldsymbol{\delta}}}_{Diff-GMM} - \hat{\hat{\boldsymbol{\delta}}}_{Sys-GMM}$. Under the null hypothesis that both estimators are consistent, the difference should be close to zero. The Hausman statistic is:

$$h = \hat{\boldsymbol{q}}' \left[ avar(\hat{\boldsymbol{q}}) \right]^{-1} \hat{\boldsymbol{q}} \overset{a}{\sim} \chi^2(K), \tag{56}$$

where $K$ is the number of parameters tested. Both approaches test whether the stronger set of orthogonality conditions is supported by the data. The incremental Sargan test evaluates the additional moments directly, while the Hausman test evaluates their implications for the estimated coefficients.

Finally, Arellano and Bond (1991) proposed a direct test for serial correlation of shocks, whose absence is crucial for the validity of instruments. In particular, the null hypothesis of the test is the absence of second order autocorrelation in the first-differenced residuals. Specifically:

$$m_2 = \frac{\widehat{\Delta \boldsymbol{v}_{-2}}' \widehat{\Delta \boldsymbol{v}_*}}{se} \overset{a}{\sim} \mathcal{N}(0,1), \tag{57}$$

where $\Delta \boldsymbol{v}_{-2}$ is the second lagged residual in differences, and $\Delta \boldsymbol{v}_*$ is the part of the vector of contemporaneous first differences for the periods that overlap with the second lagged vector. Values close to zero do not allow rejection the hypothesis of no serial correlation. Intuitively, first-differencing induces MA(1) correlation mechanically, so AR(1) in differenced residuals is expected; the crucial test is the absence of AR(2).

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